Yalun Wu

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Education

Beihang University (BUAA) *Sep 2021 – Jul 2025*
B.Eng. in Cyberspace Security

National University of Singapore (NUS) *Aug 2025 – Dec 2026, expected*
Master of Computing, NExT++ Lab
Coursework: Data Structures, Machine Learning, Deep Learning, RL **Tests:** IELTS 6.5; GRE 322, V152 / Q170

Research Interests

Trustworthy LLM systems and agent architectures: multi-step RAG, self-evolution, memory governance, runtime verification, and evaluation under strict constraints. Experience spans problem formulation, benchmark design, implementation, deployment, and operational validation.

Technical Skills

- **Agent systems:** Workflows, harnesses, task decomposition, multi-agent collaboration, tool use, structured memory, and verifiable execution.
- **RAG:** Multi-step multimodal retrieval and long-context compression; optimizing recall, evidence retention, reasoning stability, latency, and cost.
- **Trustworthy evaluation:** Benchmarks, failure taxonomies, evidence-based scoring, model comparisons, ablations, and error attribution; reproducible tests with structured outputs, schema constraints, runtime checks, and physical/safety rules.
- **Multimodal models:** VLMs, medical imaging, embodied time series, visual RAG, and business search under structural and safety constraints.
- **Engineering:** Python, PyTorch, Linux, Git, LaTeX; deployment, data pipelines, batch experiments, visualization, scientific writing, and teamwork.

Internship Experience

Meituan *May 2026 – Present*
LLM Algorithm Intern Multimodal RAG and Agents

- Built multi-step visual RAG for production knowledge-base QA and proposed the Visual Evidence Bottleneck (VEB) framework, attributing degradation to input contamination, cross-step memory decay, and limited visual capacity; a shared requirement state coordinates evidence selection, memory control, and pixel allocation, invoked adaptively by the agent as reasoning proceeds. Related work submitted to ICLR 2027.
- Raised internal image–text QA accuracy from 70.8% to 82.8% with the adaptive framework (+12.0 pp), with the best single method at 85.8%; ViDoSeek rose from 72.5% to 81.3%. The 81.5% configuration runs at 21 s P50 latency, on par with the single-step baseline.
- Transferred evidence curation and memory control to text and audio–video QA, raising accuracy on 4,547 open-ended business questions from 70.33% to 77.22%.
- Drove multimodal query rewriting to production, improving rewrite quality from 79.0% to 95.8% (+16.8 pp) at 4.01 s P90 latency.

Huazhu Group *Oct 2025 – Jan 2026*
Enterprise AI Agent Intern Food-service Supply Chain and AI Infrastructure

- Mapped centralized procurement, bulk purchasing, replenishment, and menu-planning workflows, rules, and data interfaces for supply-chain AI.
- Built multi-agent modules for procurement, product/supplier extraction, replenishment, and menu planning, linking data to recommendations.
- Consolidated store needs, prices, product constraints, and supplier policies through feedback and prototypes; explored structured skill generation and validation, informing LFUSE's scenarios and business rules.

Haitun Zhisheng Medical Technology Co., Ltd. *Apr 2025 – Sep 2025*
LLM Evaluation Algorithm Intern Vision-Language Model Evaluation

- Led ultrasound VLM deployment and benchmarking; ran experiments, verified results, and standardized outputs for model comparisons.
- Contributed to U2-BENCH and DOLPHIN through reasoning-mode comparisons, cross-anatomy generalization analysis, and ablations.
- Designed prompts, structured data, and analyzed outputs; wrote experiment and failure reports supporting paper figures and conclusions.

Beijing Feixiong Technology Co., Ltd. *Jul 2024 – Sep 2024*
Intern UAV Research and Development

- Tested UAV communication security; supported troubleshooting, fault localization, and functional verification.

Research and Projects

Agent Systems, RAG, and Business Intelligence

TAEC: Trajectory-Aware Evidence Coordination for Multi-Step Visual RAG
First author ICLR 2027, under review

- Proposed training-free trajectory-level evidence coordination for multi-step visual RAG. Shared unmet answer requirements align evidence admission, memory rendering, and visual detail allocation, reducing cross-step interference.
- Designed joint coverage and redundancy filtering, adaptive memory rendering, and requirement-aware visual budgeting to retain useful observations, compress resolved information, and prune failed branches.
- Achieved **66.8%** mean accuracy across 12 settings on 3 benchmarks and 4 backbones, **4.5 pp** above DAG; same-backbone gains on ViDoSeek, SlideVQA, and MMLongBench-Doc were **7.4/6.9/5.6 pp**.

EvolveMAS: A Unified Self-Evolution Framework for Decentralized Multi-Agent Systems
First author ICLR 2027, under review

- Proposed a self-evolution framework for decentralized multi-agent systems, unifying within-episode adaptation and cross-episode knowledge evolution across cooperative, mixed-motive, and adversarial settings.
- Paired each agent with a MemoryAgent managing episodic, semantic, and working memory outside its reasoning context; Execute–Evaluate–Evolve distills reusable strategies and partner profiles without updating model parameters.

- Achieved the highest normalized mean score of **77.2** across 5 benchmarks and 6 configurations, **7.7 points** above Self-Memory; factorial ablations isolated the contributions of memory governance and cross-episode knowledge reuse.

The Unreliable Progress Bar: Can LLM Agents Reliably Report Task Progress Throughout Execution?
Corresponding author — ICLR 2027, under review

- Agent frameworks use self-reported task progress to decide whether to continue or stop; studied whether LLMs can report progress reliably at every stage of execution, and where and how their reports fail.
- Placed reporting checkpoints across the task lifecycle on the public τ^2-bench and the controlled StageIF testbed, giving an evaluation protocol that spans the whole course of execution.
- Found reliability to be stage-dependent: nearly every deployed model is reliable only at some stages; most lose accuracy mid-task and recover once done, while the newest generation turns conservative at the finish, so frameworks should not control task flow on state reports alone.

LHMG: Long-Horizon Memory Governance for LLM Agents
Co-first author — AAAI 2027, under review

- Addressed fact conflicts and memory drift through versioned revision, risk gating, constrained decoding, and governed forgetting.
- Versioning and immutable audit logs raised conflicting-fact update accuracy from 40% to 100%; NLI-based contradiction control improved cross-turn consistency by up to 13 pp in long-horizon and multi-session tests.

LFUSE: Learned Evidence Fusion for Noisy Spoken and Image-Grounded Product Search
Co-first author — ICASSP 2027, under review

- Helped design tasks, workflows, and agent prototypes for noisy speech/image search in Huazhu's procurement, replenishment, and menu planning.
- Turned user requirements, product constraints, supplier policies, and operational feedback into search tasks and evaluation protocols.

SchemaBridge: A Unified Schema Interface and Corpus for Cross-Task Semantic Parsing
First author — NAACL 2027, planned submission

- Led unified semantic parsing and writing, mapping NER, RE, EE, and SLU to hierarchical category–args schemas for devices and business assistants.
- Unified 61 datasets and 2.75M multilingual examples; analyzed Intent Engine's Parallel Match, latency, and training–deployment schema mismatches.

Trustworthy Evaluation in Embodied and Safety-Constrained Settings

FLY-EVAL++: An Evidence-Driven Evaluation Protocol for Safety-Constrained Flight Prediction
First author — COLM 2026

- Proposed a two-stage protocol for flight prediction under physical and safety constraints: deterministic evidence-atom checks of output structure, physical feasibility, and safety error thresholds, followed by interpretable rubric aggregation without LLM-based scoring.
- Built tasks with 34 sensor channels, multi-stage attitude–trajectory prediction, and 20 safety error-threshold checks grounded in FAA and other aviation guidance; extended evaluation to multi-step rollouts to analyze accuracy degradation and safety risks as temporal complexity increases.
- Screened 66 LLMs and fully evaluated 21, revealing safety-score gaps exceeding 28 points among similarly accurate models. Identified failures in protocol compliance, field validity, physical consistency, safety coverage, and prediction accuracy.
- Demonstrated why predictive accuracy alone misses operational reliability; established a reusable process for deriving metrics from domain rules, testing safety boundaries, and analyzing failures in other embodied applications.

PilotBench: A Benchmark for General Aviation Agents with Safety Constraints
First author — IJCNN 2026

- Built joint trajectory–attitude prediction tasks from 708 real flights, including preprocessing, model evaluation, and results analysis.
- Evaluated 41 models and conventional predictors; designed PILOT-SCORE for regression accuracy, instruction following, and safety compliance, leading experiments, verification, and comparisons.
- Analyzed accuracy–controllability trade-offs and dynamic failures, laying foundations for FLY-EVAL++ evaluation and FTAP modeling.

Safety-Constrained Temporal Flight Modeling with Foundation Models
Co-first author — Nature Machine Intelligence, planned submission

- Reconstruct FTAP time series and integrate public aviation data for a high-fidelity, physics-constrained long-horizon flight prediction benchmark.
- Distinguish feasible predictions from plausible but unsafe outputs; develop safety-constrained training objectives, ablations, and validation.

Domain-Specific Large Models and Multimodal Infrastructure

U2-BENCH: Benchmarking Large Vision-Language Models on Ultrasound Understanding
Core contributor; co-author — ICLR 2026

- Helped build an ultrasound benchmark with 7,241 images, diagnostic texts, and 8 tasks spanning classification, detection, regression, and generation.
- Evaluated 20 general-purpose and medical VLMs, verifying results and standardizing outputs for comparability and reproducibility.
- Analyzed performance and failures; prepared visualizations, comparisons, and key figures to explain differences in ultrasound understanding.

H2HTalk: Evaluating Large Language Models as Emotional Companion
Core contributor; second author — NLPCC 2025 Oral Paper

- Helped build a companion benchmark (dialogue, recollection, itinerary) and its scoring; evaluated 50 models on 4,650 scenarios in 11 subtasks.
- Scored responses with BLEU, ROUGE, and BGE-M3 semantic similarity, plus GPT-4o rubrics and human review for itinerary planning; analyzed failures in memory retention, long-horizon planning, and implicit needs.
- Studied safety and emotion regulation with the attachment-theory-based Secure Attachment Persona (SAP); removing it raised violation rates in high-risk scenarios from 0.7% to 7.1%.

DOLPHIN: A Multimodal Large Language Model for Ultrasound Understanding
Core contributor; listed immediately after co-first authors — CVPR 2027, planned submission

- Evaluated post-training, instruction tuning, and UARPO reinforcement learning; compared reasoning and standard modes across 16 anatomical regions, analyzing measurement errors and missed detections.
- U2-Score: 0.5835; reasoning versus standard mode raised diagnosis/detection accuracy by 2.4%/16% and reduced measurement RMSE by 10.6%.

NeuroMark: Multi-Modal Evaluation for Neuropsychiatry with Joint Trustworthiness Metrics
Co-first author — Nature Communications, planned submission

- Help define medical-text reasoning and brain-image understanding tracks for neuropsychiatric evaluation and manuscript planning, including task scope, model coverage, and trustworthiness.
- Lead BrainVLM: integrate multi-source brain MRI; evaluate 16 VLMs on 6 tasks with unified inference, parsing, scoring, and cross-task analysis.